G10-DESIGN-AUDIT-v1: Continuation by spatial structure - backward/unique continuation, NOT new critical norm

ZFL-CX

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$$\boxed{G7 = FROZEN,\ G8 = CLOSED/BLOCKED,\ G9\ BLOCKED\ (2\ gates)}$$

$$\boxed{\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\},\ T^* < \infty}$$

1 New class

NOT: estimate W_+, NOT: blow-up profile compactness requiring $r_n^{-1}E_0$ bound, NOT: new critical norm X with $\|u\|_X \to \infty$.

New axis:

$$\boxed{\text{What spatial structure must become singular if } T^* < \infty, \text{ and is that structure impossible by rigidity?}}$$

Family: backward uniqueness, unique continuation, Carleman, Liouville with additional structure, geometric spatial condition forcing regularity. Known in NS literature (ESS, etc.) but need activation from $\mathcal{C}_{PROVEN}$ only.

2 ZFL rule

$$\boxed{\text{“backward uniqueness exists”} \neq \text{“we can apply it with our } C_i\text{”}}$$

No smuggling: $C_{PROVEN} \Rightarrow$ structure, not $C_{PROVEN} + hyp_{new}$.

3 4 gates - all must pass for G10-01

Gate A - Derivable spatial structure: Does $T^* < \infty + \mathcal{C}_{PROVEN}$ imply a spatial property usable for continuation? Candidates: - smallness in L^3 outside concentration balls? - u vanishing to infinite order at T^*? - vorticity direction ξ has some C^α patch? - u coincides with regular solution on open set? Must be derived, not assumed. If NO: $\boxed{G10 = CLOSED/BLOCKED}$.

Gate B - Exact hypotheses of continuation theorem: Identify theorem: e.g., Escauriaza-Seregin-Sverak backward uniqueness for heat operator $|\partial_t u + \Delta u| \leq C(|u| + |\nabla u|)$ with $|u| \leq Ce^{C|x|^2}$ and $u(\cdot, 0) = 0 \Rightarrow u \equiv 0$; or unique continuation for Stokes; or Carleman for NS vorticity. List precise bounds required: $u \in L_t^\infty L^q$, decay, vanishing order, differential inequality. If hypotheses require $L_t^\infty L^3$ global small or $e^{C|x|^2}$ or Type I, and we don't have it from $\mathcal{C}_{PROVEN}$, then $\boxed{BLOCKED}$.

Gate C - Transfer from our C_i: Does $\mathcal{C}_{PROVEN}$ provide the coefficients in Gate B? C_{L^3} gives global L^3 but not decay $e^{C|x|^2}$. C_F^{conc} gives $\|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0$, opposite of vanishing. C_I^{freq} gives $N_c \to \infty$, not vanishing order. C_J^{neg} gives loss of ξ regularity, not gain needed for Carleman. Check: no automatic transfer. If fails: $\boxed{G10 = CLOSED/BLOCKED}$.

Gate D - Real contradiction with $T^* < \infty$: If A+B+C pass, does theorem imply $\bar{u} \equiv 0$ or u extends beyond T^* contradicting $T^* < \infty$? Must produce $\perp$, not just regularity on open set.

4 Kill criteria

KILLED if: - A requires assuming spatial smallness/analyticity not in $\mathcal{C}_{PROVEN}$, - B requires Type I, $L_t^\infty L^3$ uniform smallness, or exponential bound not derivable, - C cannot bridge concentration (lower bound) to vanishing (upper bound) - they are opposite, - D only gives partial regularity, not full continuation beyond T^*.

5 Why this is different from G8/G9

G8: tried $norm \to W_+$ bridge - killed by stretching algebra scale and shear barrier $\bar{\alpha} \equiv 0$. G9: tried $T^* < \infty \Rightarrow u_n \to \bar{u}$ ancient - blocked by $r_n^{-1} E_0$ supercriticality and weak limit losing ε_0. G10: tries $T^* < \infty \Rightarrow$ spatial structure $\Rightarrow$ backward/unique continuation $\Rightarrow \perp$ without needing W_+ control nor u_n compactness. Genuinely new class if A holds.

$$\boxed{\text{This audit only asks: can } \mathcal{C}_{PROVEN} \text{ activate continuation/rigidity? No G10-01 yet}}$$

$$\boxed{\text{If A,B,C fail } \Rightarrow G10 = CLOSED/BLOCKED, STOP = PROGRESO}$$